Early-Life Disadvantage and Later-Life Cognitive and Mental Health in India: LASI Wave 1 (Corrected Analysis, v2)

Status: Revised report following an independent audit of the original (v1) analysis. v1 is archived at `report/final_research_report.md`. All deviations from the original protocol are listed in `protocol.md`, Section 12 (Protocol Amendment 1).

> **Interpretation guardrail.** LASI Wave 1 is cross-sectional. All findings are *associations* (prevalence ratios, mean differences). The data do not permit causal or mediation claims, and they cannot show within-person change.

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1. Why this revision exists

An audit of v1 found problems that changed conclusions. This table lists each one and the corrective action.

Audit finding	v1 behaviour	v2 correction
Cognitive score was a schooling proxy	Items skipped by unschooled respondents scored 0 (`fillna(0)`)	Primary outcome = memory + orientation (0–28); skip-aware full battery kept as sensitivity
Orientation partial credit	Missing orientation items counted as 0	All 8 items required
Cut-off	Unweighted pooled 10th percentile	Survey-weighted 10th percentile; school-stratum-specific version as sensitivity
Interaction was not tested	Visual reading of one covariate profile	Joint Wald F test; standardized predictions averaged over each age group's covariates
Survey design	Strata ignored	Stratified-cluster linearization (72 strata, 2,428 clusters)
Model comparability	Different samples across M0–M3	One common complete-case sample per outcome
Regional confounding	No state control	State fixed effects from Model 1
"Independent" claims	Exposures modelled one at a time	Mutually adjusted models and a cumulative index
Informative missingness	Complete-case only	IPW comparison
Causal language	"Mediated", "scarring", "HPA programming"	Associational wording only
SES contrast bug (found during v2 coding)	not applicable	SES always modelled as a joint 3-level variable (reference = well-off)

2. Primer on Ageing, Epidemiological, and Survey Methodology (Layman's Glossary)

To ensure that the empirical findings of this report are accessible to non-specialists, policymakers, and interdisciplinary readers, this section defines the essential technical terms from life-course epidemiology, gerontology, and survey methodology. Common concepts have been omitted; only complex, domain-specific terms are clarified below.

2.1 Life-Course Epidemiology and Ageing Paradigms

- **Biological Scarring (Sensitive / Critical Period Model):** The biological principle that severe physical shocks, malnutrition, or infectious illnesses during early childhood permanently alter the body's physiological set-points (such as chronic immune inflammation, vascular stiffness, or neuroendocrine stress regulation). These physiological alterations remain dormant or active throughout adult life, re-emerging

decades later in old age as elevated risks of depression, disability, or physical frailty, even if the individual achieved upward socioeconomic mobility in adulthood.

- **Accumulation of Disadvantage (Cumulative Adversity):** A developmental trajectory whereby initial childhood disadvantage triggers a cascade of sequential hardships across the lifespan. Being born into poverty frequently constrains schooling opportunities, channeling individuals into hazardous manual labor with low wages, substandard living conditions, poorer nutrition, and delayed medical care. Over 50 to 70 years, these recurring hardships accumulate, gradually degrading physical and cognitive reserves.
- **Selective Survival (Mortality Selection / Survivor Bias):** A statistical phenomenon observed in populations with historically high mortality rates. In mid-20th-century India, infants and children born into severe poverty faced substantial risks of premature death. Consequently, impoverished individuals who successfully survived into late adulthood (past age 75) often possess exceptional biological hardiness, psychological resilience, or favorable genetic profiles. In cross-sectional surveys, this can cause the oldest-old disadvantaged survivors to appear as healthy as---or even healthier than---their wealthier peers, because the most vulnerable individuals died prior to study enrollment.
- **Cross-Sectional Baseline (Non-Causal Guardrail):** A single snapshot observation of a population at one point in time. Because LASI Wave 1 was collected cross-sectionally, we observe older adults at different ages simultaneously rather than tracking the same individual across their life. Therefore, statistical findings represent *cross-sectional associations* across cohort strata and cannot establish individual disease progression, longitudinal cognitive decline, or definitive cause-and-effect relationships.

2.2 Survey Sampling and Statistical Design

- **Primary Sampling Unit (PSU) and Secondary Sampling Unit (SSUID):** In massive national demographic surveys, interviewers cannot sample individuals randomly from a national registry. Instead, multi-stage cluster sampling is employed: first, geographic sub-districts (taluks or tehsils) are selected as Primary Sampling Units (PSUs); within those districts, specific rural villages or urban census enumeration blocks are selected as Secondary Sampling Units (SSUIDs). All households within sampled clusters are then enumerated.
- **Stratification (72 Strata):** The technique of dividing the total population into non-overlapping geographic subgroups (called "strata") before sampling. In LASI, 72 strata were created by crossing each of the 36 States and Union Territories with sector of residence (Rural vs. Urban). Stratification guarantees that the sample accurately represents every geographic and regional corner of India, preventing rural or remote regions from being overlooked.
- **Sampling Weights (`indiaindividualweight`):** To ensure reliable regional estimates, smaller states (such as Goa or Sikkim) were sampled at a higher rate relative to their population size than populous states (such as Uttar Pradesh). Sampling weights mathematically rescale each respondent's contribution so that one participant in an oversampled region represents fewer citizens and one in an undersampled region represents more citizens. Applying weights ensures that calculated statistics accurately reflect the national population of India.
- **Taylor Series Linearization (Cluster-Robust Standard Errors):** Households located in the same village share identical environmental risks, drinking water sources, local clinic infrastructure, and socioeconomic conditions. Because their health outcomes are mathematically correlated rather than independent, standard textbook formulas would underestimate uncertainty (yielding falsely narrow confidence intervals). Taylor series linearization calculates cluster-robust variance estimates by measuring variation between clusters centered within each stratum.
- **Survey Degrees of Freedom (df = 2,356):** In complex survey analysis, the degrees of freedom for statistical testing is determined not by the number of individual respondents (N = 66,606), but by the number of independent sampling clusters minus the number of strata (2,428 clusters - 72 strata = 2,356).
- **Proxy Interview (`rproxy`):** An interview protocol used when an older respondent was cognitively incapacitated, speech-impaired, severely ill, or deaf. An informed family member (proxy) answered basic demographic and health questions on the respondent's behalf. Because family proxies cannot complete direct cognitive performance tasks, proxy respondents (N = 705) were systematically excluded from direct cognitive assessment modules.

2.3 Cognitive Assessment and Mental Health Metrics

- **CES-D-10 Scale (Center for Epidemiologic Studies Depression):** A standardized 10-item psychiatric screening tool evaluating depressive symptom frequency during the past week (e.g., feelings of sadness, loneliness, fatigue, disrupted sleep, and hopelessness). Responses are aggregated into a score from 0 to 10; scores of ≥ 4.0 are clinically validated as indicating significant depressive symptoms.
- **Cognitive Battery and Item-Skip Bias:** A multi-domain assessment testing memory, temporal/spatial orientation, numeracy, executive function, and object naming. In developing nations with high illiteracy, respondents who never attended school are protocol-skipped on formal arithmetic tasks (e.g., serial 7 subtractions or backward counting). If skipped tasks are improperly coded as "zero points" (failure), the test inadvertently measures lack of formal schooling rather than neurological impairment.

- **Orientation vs. Word Recall Domains:** Orientation tests fundamental cognitive awareness of time and place (identifying the day, month, year, state, district, and local landmarks). Word recall tests episodic memory by reading a list of 10 common nouns to the respondent and recording immediate recall and delayed recall (after a short interference task).

2.4 Analytical Models and Epidemiological Estimands

- **Modified Poisson Regression vs. Odds Ratios (OR):** In medical studies, logistic regression is commonly used to produce Odds Ratios. However, when an outcome is common (such as depressive symptoms, which affect 28.2% of older Indians), Odds Ratios severely exaggerate relative risk (transforming a true 20% increase into a misleading 50% increase). Modified Poisson regression with robust standard errors estimates the **Prevalence Ratio (PR)**, which directly conveys the true relative percentage difference between groups.
- **Average Marginal Effects / Standardized Predictions (AME):** Instead of calculating predicted probabilities for a hypothetical "average person" (who does not realistically exist), AME takes every real individual in the survey sample, computes their predicted health probability under contrasting childhood conditions while preserving their actual age, sex, and comorbidities, and averages the results across the full population.
- **Inverse-Probability Weighting (IPW):** A statistical method designed to correct for potential selection bias caused by missing cognitive tests. Because frail, very old (75+), and unschooled respondents were more likely to have missing cognitive assessments, IPW statistically upweights the complete tests of similar respondents, mathematically restoring sample balance under the Missing at Random (MAR) assumption.
- **Monthly Per Capita Consumption Expenditure (MPCE):** The gold standard measure of economic status in Indian household surveys. In developing, largely rural economies, asking about formal annual income or wealth is notoriously unreliable due to informal labor and agricultural seasonality. MPCE measures total household spending on food, fuel, health, and consumer goods divided by household size.

3. Data and Empirical Methodology

- **Source:** LASI Wave 1 (2017–18), individual file; adults aged ≥45 (N = 66,606).
- **Design variables:** national individual weight, 2,439 `ssuid` clusters, 72 strata (state × residence). `ssuid` is a secondary sampling unit, and no PSU identifier exists in the release, so variances may be somewhat understated.
- **Exposures (retrospective, up to age 16):** childhood SES (`ht235`; "Varied" excluded from the main analysis), childhood health (`ht231`; Fair/Poor/Very poor vs Good/Very good), bedridden ≥1 month (`ht233`).
- **Outcomes:** (i) memory (immediate + delayed recall, 0–20) plus orientation (8 items), 0–28, with a binary indicator for the lowest weighted decile (≤11 points; 12.8% weighted prevalence); (ii) CES-D-10 ≥4 (28.2% weighted).
- **Covariates:** M1 = age, sex, residence, caste, marital status, state; M2 = M1 + education + MPCE quintile (primary); M3 = M2 + chronic-condition count + self-rated health (pathway/exploratory).
- **Estimation:** weighted modified Poisson (PR) for binary outcomes, weighted linear models for scores. Design-based confidence intervals use t with 2,356 degrees of freedom. Common complete-case samples: cognition N = 57,245; depression N = 62,047.

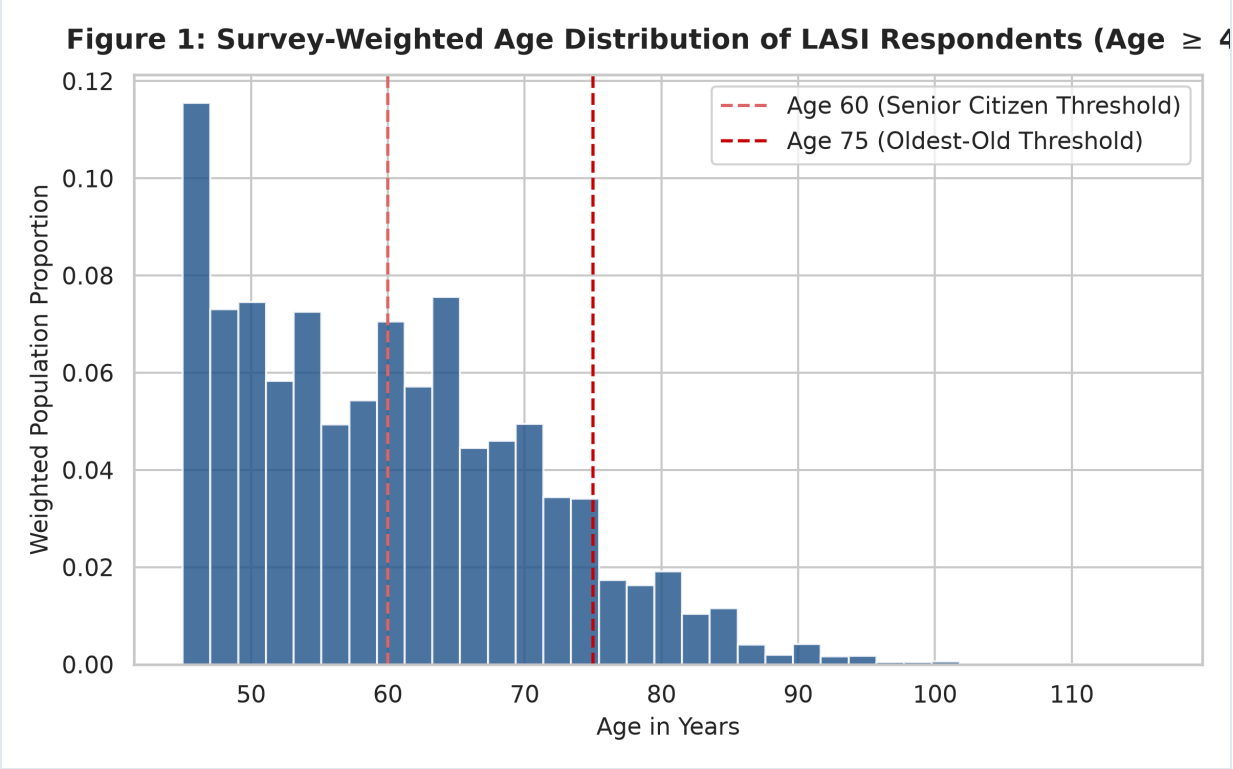


Figure 1: Survey-weighted age distribution of adults aged 45+ (N = 66,606).

Table 1. Sample characteristics (v1 descriptive table, unchanged; survey-weighted)

Characteristic	Category	Unweighted_N	Weighted_Pct
Age Group	45-59	34704	49.77
Age Group	60-74	25022	38.8
Age Group	75+	6880	11.42
Sex	Female	35567	54.06
Sex	Male	31039	45.94
Place of Residence	Rural	43240	68.53
Place of Residence	Urban	23366	31.47
Social Group / Caste	General/None	16295	24.84
Social Group / Caste	OBC	25212	46.66
Social Group / Caste	SC	11014	19.66
Social Group / Caste	ST	11746	8.84
Marital Status	Divorced/Separated	1006	1.23
Marital Status	Married/Partnered	49949	73.86
Marital Status	Never Married	868	1.19
Marital Status	Widowed	14780	23.72
Educational Attainment	Graduate and above	3429	5.19
Educational Attainment	Higher secondary/Diploma	3175	4.94
Educational Attainment	Less than primary	7598	10.92
Educational Attainment	Middle completed	6337	8.35
Educational Attainment	No formal education	31350	50.62
Educational Attainment	Primary completed	8761	12.05
Educational Attainment	Secondary completed	5953	7.93
Household MPCE Quintile	1	13181	20.86
Household MPCE Quintile	2	13403	21.24
Household MPCE Quintile	3	13371	20.49
Household MPCE Quintile	4	13412	19.42
Household MPCE Quintile	5	13239	17.99
Childhood SES	Average	35438	49.32
Childhood SES	Poor	25605	42.16
Childhood SES	Pretty well-off	4848	7.77
Childhood SES	Varied	394	0.76
Childhood Health	Good	58272	87.46
Childhood Health	Poor	7369	12.54
Childhood Bedridden	No	62881	94.25
Childhood Bedridden	Yes	3370	5.75
Cumulative Disadvantage Score	0.0	34506	48.19
Cumulative Disadvantage Score	1.0	26556	44.06
Cumulative Disadvantage Score	2.0	4003	6.86
Cumulative Disadvantage Score	3.0	466	0.88

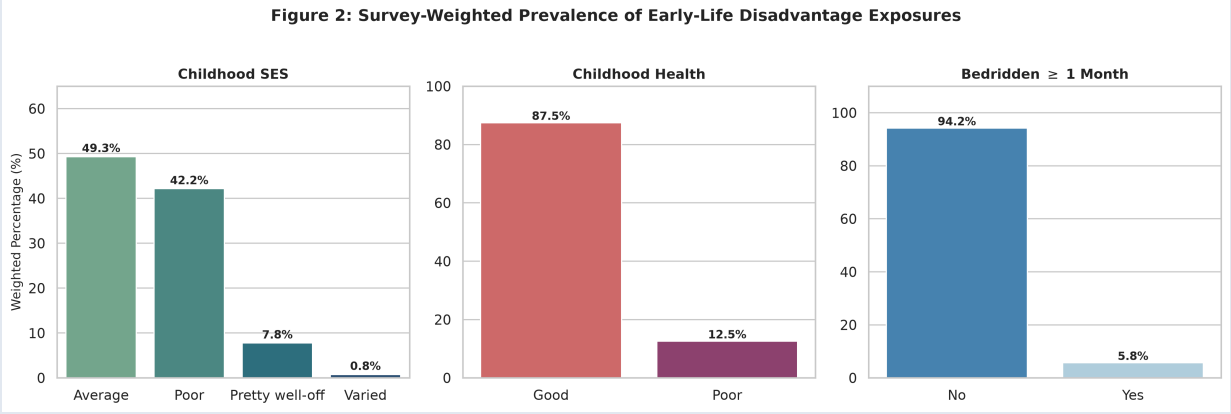


Figure 2: Survey-weighted prevalence of childhood disadvantage exposures.

Table 2 (v2). Outcome prevalence and means by age group (design-based 95% CIs)

Age_group	Outcome	Estimate	95% CI	N
45-59	Cognitive impairment (%)	7.61	7.0–8.22	32301
45-59	Depressive symptoms (%)	26.28	24.99–27.58	34009
45-59	Mem+orient score (mean, 0-28)	17.01	16.85–17.17	32301
45-59	CES-D score (mean)	2.86	2.81–2.91	34009
60-74	Cognitive impairment (%)	15.61	14.58–16.65	22223
60-74	Depressive symptoms (%)	29.1	27.8–30.41	24370
60-74	Mem+orient score (mean, 0-28)	15.51	15.38–15.65	22223
60-74	CES-D score (mean)	3.01	2.95–3.06	24370
75+	Cognitive impairment (%)	28.85	26.2–31.5	5257
75+	Depressive symptoms (%)	34.18	31.02–37.33	6446
75+	Mem+orient score (mean, 0-28)	13.88	13.65–14.11	5257
75+	CES-D score (mean)	3.27	3.15–3.39	6446
All	Cognitive impairment (%)	12.79	12.03–13.54	59781
All	Depressive symptoms (%)	28.24	27.19–29.28	64825
All	Mem+orient score (mean, 0-28)	16.13	16.0–16.25	59781
All	CES-D score (mean)	2.96	2.92–3.0	64825

4. What went wrong with the cognitive outcome, and what the fix did and did not do

In the v1 composite, impairment was classified in 21% of the unschooled and 0.1% of graduates. The v2 primary outcome still shows a steep schooling gradient, because recall and orientation also depend on education.

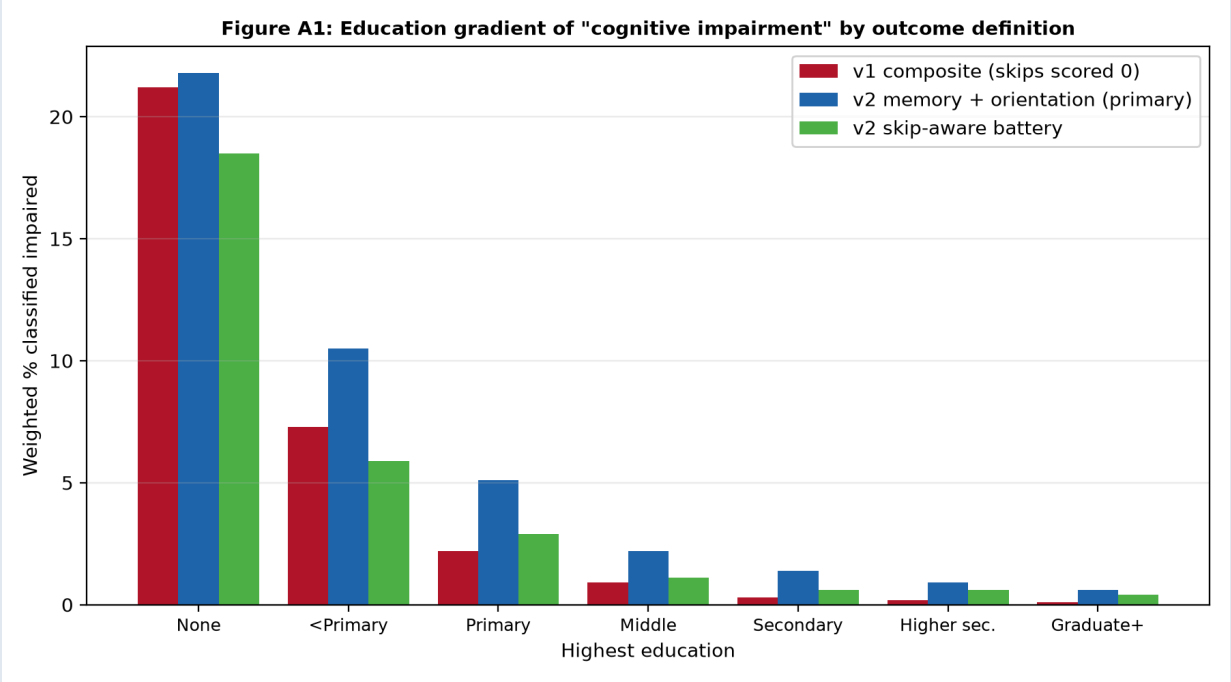


Figure A1: Education gradient of classified cognitive impairment by outcome definition.

Table S5. Weighted % classified impaired, by education and outcome definition

Education	v1_composite_impaired_pct	v2_mem_orient_impaired_pct	v2_battery_impaired_pct	n_v2
No formal education	21.2	21.8	18.5	26760
Less than primary	7.3	10.5	5.9	6829
Primary completed	2.2	5.1	2.9	8147
Middle completed	0.9	2.2	1.1	5993
Secondary completed	0.3	1.4	0.6	5681
Higher secondary/Diploma	0.2	0.9	0.6	3056
Graduate and above	0.1	0.6	0.4	3314

Take-away: no absolute-threshold cognitive measure is free of education. Education-stratified analysis and the school-stratum-specific cut-off are therefore essential, not optional. The Model 2 attenuation below cannot be read as a "pathway" through education.

5. Primary Results: Cognition

Table 3 (v2). Poor vs well-off childhood SES and other exposures with cognitive impairment (PR, 95% CI; common sample)

Exposure	Model 0	Model 1	Model 2	Model 3
Childhood SES: Average	0.97 (0.82–1.15) p=0.711	0.98 (0.84–1.14) p=0.777	0.80 (0.70–0.91) p<.001	0.80 (0.71–0.91) p<.001
Childhood SES: Poor	1.61 (1.33–1.96) p<.001	1.36 (1.16–1.59) p<.001	0.93 (0.82–1.05) p=0.241	0.91 (0.81–1.03) p=0.148
Childhood bedridden ≥1 month	0.90 (0.77–1.06) p=0.222	0.94 (0.81–1.09) p=0.404	1.00 (0.86–1.18) p=0.947	0.99 (0.84–1.16) p=0.868
Childhood health: Fair/Poor/Very poor	0.97 (0.81–1.17) p=0.773	0.97 (0.85–1.12) p=0.719	0.95 (0.82–1.10) p=0.487	0.95 (0.82–1.09) p=0.458

Table 3b. Mean difference in memory + orientation score (points, 0–28)

Exposure	Model 0	Model 1	Model 2	Model 3
Childhood SES: Poor	-1.65 (-1.95–-1.34)	-1.28 (-1.54–-1.01)	-0.32 (-0.56–-0.09)	-0.24 (-0.48–-0.01)
Childhood health: Fair/Poor/Very poor	-0.02 (-0.43–0.39)	-0.05 (-0.29–0.18)	0.01 (-0.23–0.26)	0.04 (-0.19–0.27)

Findings

- **Childhood SES.** Poor vs well-off: PR 1.61 (1.33–1.96) unadjusted and 1.36 (1.16–1.59) after demographics and state. After adult education and MPCE the association is not distinguishable from 1 (PR 0.93, 0.82–1.05). On the continuous score, a small residual deficit remains (−0.32 points, −0.56 to −0.09).
- **"Average" childhood SES** has a *lower* prevalence than well-off after adult SES adjustment (PR 0.80, 0.70–0.91). The well-off group's relatively high prevalence is unexplained here. Possible reasons are selective survival, differences in recall and reporting, or residual confounding; none of these was tested.
- **Childhood health and bedridden history** show no association with memory + orientation impairment in any model (PRs 0.90–1.00; all CIs include 1).
- **Mutual adjustment** (Table 6): the pattern is the same. Poor SES PR 1.36 (1.15–1.60) in M1 becomes 0.93 (0.82–1.06) in M2. The cumulative index gives PR 1.05 per adversity (0.995–1.12).

Education-stratified and outcome-definition analyses (Table S4a/S4b)

Analysis	Outcome	Term	Estimate (95% CI)	N
S4b Schooling stratum: Never attended school	Cognitive impairment (primary)	ses_poor	0.91 (0.79–1.05) p=0.188	25684
S4b Schooling stratum: Never attended school	Cognitive impairment (primary)	hlth_poor	0.97 (0.83–1.15) p=0.745	25684
S4b Schooling stratum: Never attended school	Cognitive impairment (school-stratum decile)	ses_poor	0.92 (0.73–1.15) p=0.448	25684
S4b Schooling stratum: Never attended school	Cognitive impairment (school-stratum decile)	hlth_poor	1.01 (0.83–1.23) p=0.916	25684
S4b Schooling stratum: Never attended school	Mem+orient continuous (beta)	ses_poor	-0.39 (-0.72–-0.05) p=0.025	25684
S4b Schooling stratum: Any schooling	Cognitive impairment (primary)	ses_poor	0.99 (0.76–1.29) p=0.950	31561
S4b Schooling stratum: Any schooling	Cognitive impairment (primary)	hlth_poor	0.81 (0.59–1.12) p=0.209	31561
S4b Schooling stratum: Any schooling	Cognitive impairment (school-stratum decile)	ses_poor	1.01 (0.82–1.23) p=0.940	31561
S4b Schooling stratum: Any schooling	Cognitive impairment (school-stratum decile)	hlth_poor	0.94 (0.78–1.14) p=0.525	31561
S4b Schooling stratum: Any schooling	Mem+orient continuous (beta)	ses_poor	-0.18 (-0.47–0.10) p=0.208	31561

Among respondents who never attended school, poor childhood SES is not associated with impairment (PR 0.91, 0.79–1.05), but the continuous score shows a small deficit (−0.39 points, −0.73 to −0.05). Among those with any schooling, there is no association.

Analysis	Outcome	Term	Estimate (95% CI)	N
S4a Alt. cognitive outcome / Model 1: Demographics + state	Primary: lowest weighted decile mem+orient	ses_poor	1.36 (1.16–1.59) p=0.000	57245
S4a Alt. cognitive outcome / Model 1: Demographics + state	Primary: lowest weighted decile mem+orient	hlth_poor	0.97 (0.85–1.12) p=0.719	57245
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Primary: lowest weighted decile mem+orient	ses_poor	0.93 (0.82–1.05) p=0.241	57245
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Primary: lowest weighted decile mem+orient	hlth_poor	0.95 (0.82–1.10) p=0.487	57245
S4a Alt. cognitive outcome / Model 1: Demographics + state	Skip-aware full battery (% of max), lowest decile	ses_poor	1.97 (1.60–2.42) p=0.000	57245
S4a Alt. cognitive outcome / Model 1: Demographics + state	Skip-aware full battery (% of max), lowest decile	hlth_poor	1.25 (1.11–1.41) p=0.000	57245
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Skip-aware full battery (% of max), lowest decile	ses_poor	1.29 (1.07–1.57) p=0.009	57245
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Skip-aware full battery (% of max), lowest decile	hlth_poor	1.22 (1.08–1.37) p=0.001	57245
S4a Alt. cognitive outcome / Model 1: Demographics + state	Mem+orient, schooling-stratum-specific decile	ses_poor	0.90 (0.78–1.04) p=0.171	57245
S4a Alt. cognitive outcome / Model 1: Demographics + state	Mem+orient, schooling-stratum-specific decile	hlth_poor	0.98 (0.85–1.14) p=0.794	57245
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Mem+orient, schooling-stratum-specific decile	ses_poor	1.00 (0.86–1.17) p=0.991	57245
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Mem+orient, schooling-stratum-specific decile	hlth_poor	0.98 (0.85–1.13) p=0.812	57245
S4a Alt. cognitive outcome / Model 1: Demographics + state	Original v1 composite (skips scored 0)	ses_poor	1.65 (1.34–2.04) p=0.000	58202
S4a Alt. cognitive outcome / Model 1: Demographics + state	Original v1 composite (skips scored 0)	hlth_poor	1.13 (0.97–1.31) p=0.126	58202
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Original v1 composite (skips scored 0)	ses_poor	1.04 (0.89–1.22) p=0.595	58202
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Original v1 composite (skips scored 0)	hlth_poor	1.09 (0.92–1.29) p=0.310	58202
S4a Alt. cognitive outcome / Model 1: Demographics + state	Mem+orient continuous (beta, points)	ses_poor	-1.28 (-1.54–-1.01) p=0.000	57245
S4a Alt. cognitive outcome / Model 1: Demographics + state	Mem+orient continuous (beta, points)	hlth_poor	-0.05 (-0.29–0.18) p=0.656	57245
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Mem+orient continuous (beta, points)	ses_poor	-0.32 (-0.56–-0.09) p=0.007	57245
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Mem+orient continuous (beta, points)	hlth_poor	0.01 (-0.23–0.26) p=0.933	57245
S4a Alt. cognitive outcome / Model 1: Demographics + state	Skip-aware battery continuous (beta, % of max)	ses_poor	-5.33 (-6.40–-4.25) p=0.000	57245
S4a Alt. cognitive outcome / Model 1: Demographics + state	Skip-aware battery continuous (beta, % of max)	hlth_poor	-1.55 (-2.32–-0.77) p=0.000	57245
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Skip-aware battery continuous (beta, % of max)	ses_poor	-1.75 (-2.49–-1.01) p=0.000	57245
S4a Alt. cognitive outcome / Model 2: + Adult SES (Primary)	Skip-aware battery continuous (beta, % of max)	hlth_poor	-1.29 (-2.09–-0.49) p=0.002	57245

Result that complicates the story. On the *skip-aware full battery*, which adds arithmetic, executive and naming domains, poor childhood SES (PR 1.29, 1.07–1.57) and poor childhood health (PR 1.22, 1.08–1.37) remain associated with impairment after education and MPCE adjustment. With memory + orientation only, they do not. The cognition conclusion therefore **depends on how cognition is measured**. We do not claim to know which measure is more valid. Extra domains may capture executive or numeracy function that is more sensitive to early disadvantage. They may also carry literacy, vision or motor-skill effects that education adjustment does not fully remove.

Missingness and IPW

Cognition scores are missing for 7.8% of direct interviewees overall, and the missingness is selective:

Variable	Level	N	Pct_missing_cognition
age_group	45-59	34566	6.6
age_group	60-74	24772	10.3
age_group	75+	6563	19.9
education	Graduate and above	3414	2.9
education	Higher secondary/Diploma	3159	3.3
education	Less than primary	7526	9.3
education	Middle completed	6301	4.9
education	No formal education	30873	13.3
education	Primary completed	8710	6.5
education	Secondary completed	5916	4.0
childhood_ses	Average	35098	7.3
childhood_ses	Poor	25309	11.2
childhood_ses	Pretty well-off	4808	8.7
childhood_ses	Varied	389	10.3
sex	Female	35191	10.7
sex	Male	30710	7.7

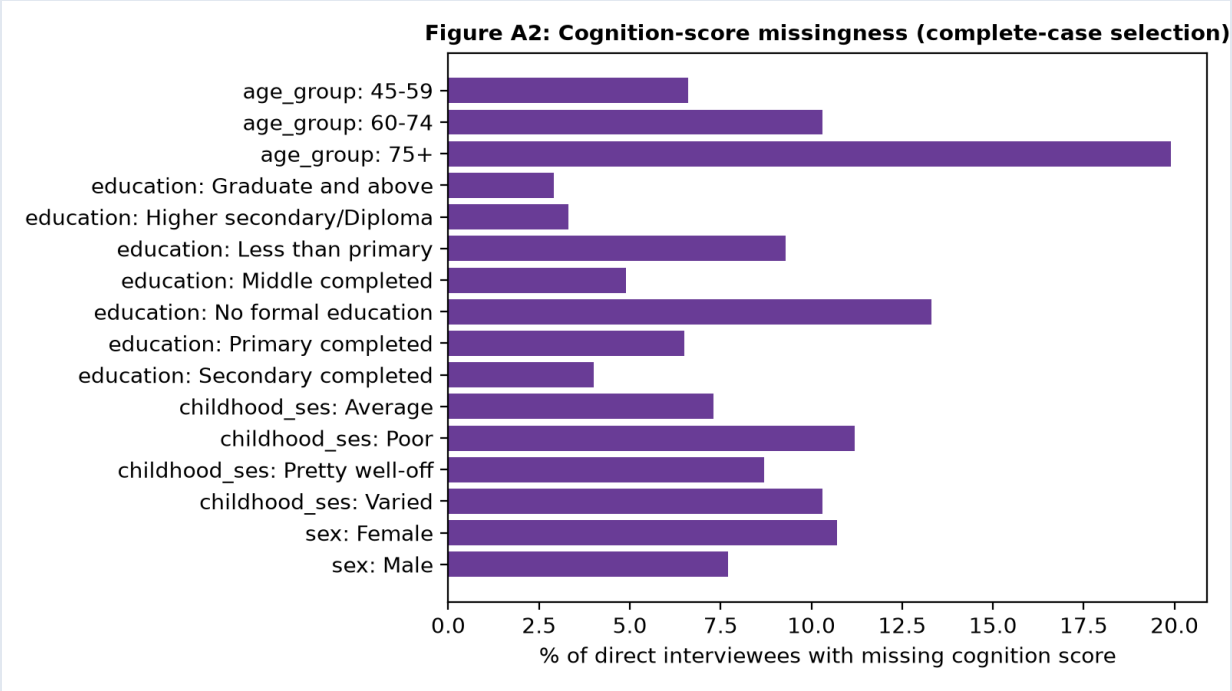


Figure A2: Missingness of cognition scores by subgroup.

IPW using covariates, exposures and CES-D availability left the estimates essentially unchanged (Poor SES, M2: 0.928 both ways):

Exposure	Complete-case	IPW-adjusted
Childhood SES: Average (vs well-off)	0.80 (0.70–0.91)	0.80 (0.71–0.90)
Childhood SES: Poor (vs well-off)	0.93 (0.82–1.05)	0.93 (0.82–1.05)
Childhood SES: Poor (vs well-off) [joint SES model]	0.93 (0.82–1.05)	0.93 (0.82–1.05)
Childhood bedridden ≥1 month (vs No)	1.00 (0.86–1.18)	1.00 (0.86–1.17)
Childhood health: Fair/Poor/Very poor (vs Good/Very good)	0.95 (0.82–1.10)	0.94 (0.81–1.09)

This addresses missingness only under a missing-at-random assumption given the model covariates. It cannot address missingness that depends on unmeasured cognitive status.

6. Primary Results: Depressive Symptoms

Table 4 (v2). Depressive symptoms (CES-D ≥4): PR, 95% CI, common sample

Exposure	Model 0	Model 1	Model 2	Model 3
Childhood SES: Average	1.10 (0.94–1.30) p=0.233	1.20 (1.01–1.44) p=0.039	1.15 (0.99–1.33) p=0.072	1.14 (0.99–1.32) p=0.071
Childhood SES: Poor	1.17 (1.00–1.37) p=0.044	1.31 (1.09–1.58) p=0.004	1.21 (1.03–1.42) p=0.019	1.15 (0.99–1.34) p=0.068
Childhood bedridden ≥1 month	1.14 (1.01–1.29) p=0.037	1.14 (1.01–1.27) p=0.034	1.16 (1.05–1.28) p=0.004	1.11 (1.01–1.23) p=0.029
Childhood health: Fair/Poor/Very poor	1.33 (1.20–1.47) p<.001	1.21 (1.08–1.36) p=0.002	1.20 (1.07–1.34) p=0.002	1.19 (1.06–1.33) p=0.002

Findings

- **Childhood health:** poor vs good is associated with a 33% higher prevalence unadjusted (PR 1.33, 1.20–1.47) and 20% higher after adult SES (PR 1.20, 1.07–1.34). v1 reported 1.31 for Model 2; the difference comes mainly from the common sample, state fixed effects and the design-based variance.
- **Childhood SES:** poor vs well-off, PR 1.21 (1.03–1.42) in M2, falling to 1.15 (0.99–1.34) once adult health is added.
- **Bedridden ≥1 month:** PR 1.16 (1.05–1.28) in M2 and 1.11 (1.01–1.23) in M3.
- **Cumulative index:** each additional childhood adversity is associated with a PR of 1.11 (1.07–1.16).
- **Mutual adjustment (Table 6):** in M2, poor SES 1.19 (1.02–1.38), poor health 1.18 (1.05–1.32) and bedridden 1.12 (1.00–1.26) are each associated with depression, so the three are partly separable. In M3, only childhood health keeps a CI excluding 1 (1.17, 1.05–1.32).
- **Share attenuated by adult SES:** the Poor-SES log-PR falls by about 30% between M1 and M2. The association with childhood health attenuates little (1.21 → 1.20).

Outcome	Model	Exposure	PR (95% CI)	N
Cognitive impairment	Model 1: Demographics + state	Childhood SES: Average	0.98 (0.84–1.14) p=0.792	57245
Cognitive impairment	Model 1: Demographics + state	Childhood SES: Poor	1.36 (1.15–1.60) p=0.000	57245
Cognitive impairment	Model 1: Demographics + state	Childhood health: Fair/Poor/Very poor	0.97 (0.84–1.12) p=0.681	57245
Cognitive impairment	Model 1: Demographics + state	Childhood bedridden ≥1 month	0.94 (0.81–1.10) p=0.430	57245
Cognitive impairment	Model 2: + Adult SES (Primary)	Childhood SES: Average	0.80 (0.70–0.91) p=0.001	57245
Cognitive impairment	Model 2: + Adult SES (Primary)	Childhood SES: Poor	0.93 (0.82–1.06) p=0.277	57245
Cognitive impairment	Model 2: + Adult SES (Primary)	Childhood health: Fair/Poor/Very poor	0.95 (0.82–1.10) p=0.533	57245
Cognitive impairment	Model 2: + Adult SES (Primary)	Childhood bedridden ≥1 month	1.01 (0.86–1.18) p=0.941	57245
Cognitive impairment	Model 3: + Adult health (Pathway)	Childhood SES: Average	0.81 (0.71–0.92) p=0.001	57245
Cognitive impairment	Model 3: + Adult health (Pathway)	Childhood SES: Poor	0.91 (0.81–1.04) p=0.171	57245
Cognitive impairment	Model 3: + Adult health (Pathway)	Childhood health: Fair/Poor/Very poor	0.95 (0.83–1.10) p=0.514	57245
Cognitive impairment	Model 3: + Adult health (Pathway)	Childhood bedridden ≥1 month	0.99 (0.84–1.17) p=0.892	57245
Cognitive impairment	Model 2: + Adult SES (Primary) [cumulative index 0-3]	Per additional childhood adversity (Poor SES + poor health + bedridden)	1.05 (0.99–1.12) p=0.071	57245
Depressive symptoms	Model 1: Demographics + state	Childhood SES: Average	1.18 (1.00–1.39) p=0.044	62047
Depressive symptoms	Model 1: Demographics + state	Childhood SES: Poor	1.29 (1.08–1.54) p=0.005	62047
Depressive symptoms	Model 1: Demographics + state	Childhood health: Fair/Poor/Very poor	1.19 (1.05–1.34) p=0.005	62047
Depressive symptoms	Model 1: Demographics + state	Childhood bedridden ≥1 month	1.10 (0.97–1.25) p=0.147	62047
Depressive symptoms	Model 2: + Adult SES (Primary)	Childhood SES: Average	1.13 (0.98–1.29) p=0.090	62047
Depressive symptoms	Model 2: + Adult SES (Primary)	Childhood SES: Poor	1.19 (1.02–1.38) p=0.026	62047
Depressive symptoms	Model 2: + Adult SES (Primary)	Childhood health: Fair/Poor/Very poor	1.18 (1.05–1.32) p=0.006	62047
Depressive symptoms	Model 2: + Adult SES (Primary)	Childhood bedridden ≥1 month	1.12 (1.00–1.26) p=0.044	62047
Depressive symptoms	Model 3: + Adult health (Pathway)	Childhood SES: Average	1.12 (0.98–1.29) p=0.087	62047
Depressive symptoms	Model 3: + Adult health (Pathway)	Childhood SES: Poor	1.13 (0.98–1.31) p=0.089	62047
Depressive symptoms	Model 3: + Adult health (Pathway)	Childhood health: Fair/Poor/Very poor	1.17 (1.05–1.31) p=0.006	62047

Outcome	Model	Exposure	PR (95% CI)	N
Depressive symptoms	Model 3: + Adult health (Pathway)	Childhood bedridden ≥1 month	1.08 (0.97–1.21) p=0.157	62047
Depressive symptoms	Model 2: + Adult SES (Primary) [cumulative index 0-3]	Per additional childhood adversity (Poor SES + poor health + bedridden)	1.11 (1.07–1.16) p=0.000	62047

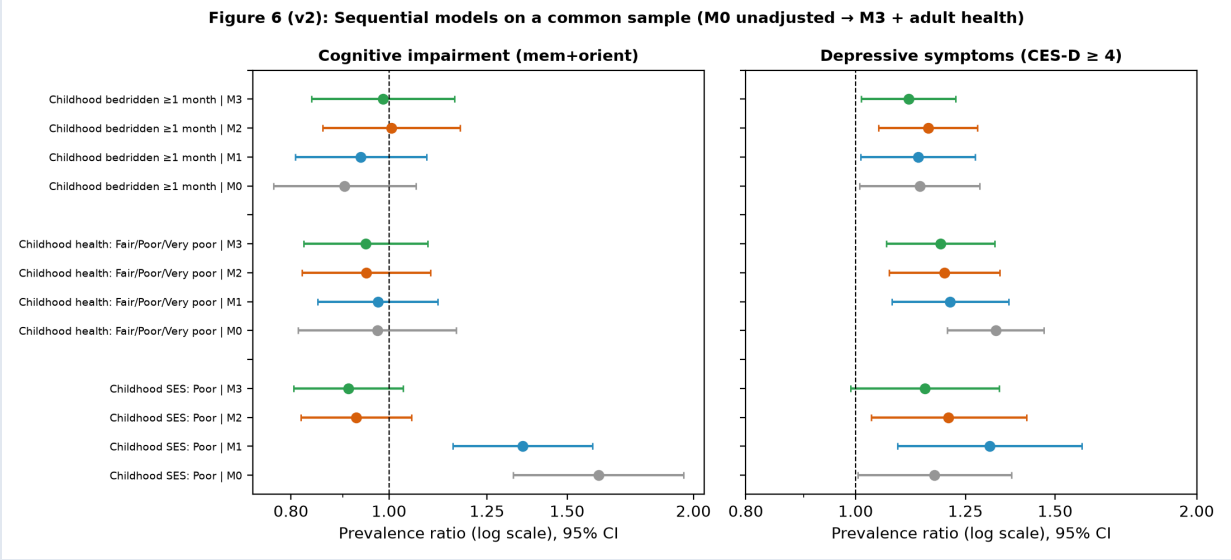


Figure 6 (v2): Sequential models M0–M3 on a common sample.

7. Age Heterogeneity (Hypothesis 4)

Table 5a (v2). Joint Wald F tests of exposure × age-group interaction

Outcome	Exposure	joint_F	df_num	joint_p	N
Cognitive impairment	Childhood SES	2.439	4	0.0451	57245
Cognitive impairment	Childhood health	0.322	2	0.7249	57245
Cognitive impairment	Childhood bedridden	0.271	2	0.7628	57245
Depressive symptoms	Childhood SES	0.34	4	0.8513	62047
Depressive symptoms	Childhood health	0.306	2	0.7361	62047
Depressive symptoms	Childhood bedridden	0.021	2	0.9795	62047

Only one of six tests (SES × age for cognition, p = 0.045) is below 0.05. It would not survive a Bonferroni correction (0.05/6 = 0.008). All other joint tests have p > 0.7.

Table 5b. Standardized prevalence (%) by childhood SES and age group

Outcome	age_group	Pretty well-off	Average	Poor
Cognitive impairment	45-59	8.16 (6.7–10.13)	6.38 (5.79–7.19)	8.45 (7.75–9.35)
Cognitive impairment	60-74	16.86 (14.52–20.2)	14.2 (13.26–15.56)	16.16 (15.2–17.29)
Cognitive impairment	75+	36.87 (30.9–43.95)	27.48 (24.99–31.02)	28.04 (26.12–31.36)
Depressive symptoms	45-59	22.69 (20.01–25.66)	26.08 (24.68–27.66)	27.01 (25.38–29.04)
Depressive symptoms	60-74	24.13 (19.66–29.45)	28.42 (27.14–30.34)	30.33 (28.83–32.23)
Depressive symptoms	75+	31.58 (23.9–42.81)	33.0 (29.66–36.8)	35.56 (32.19–39.99)

Table 5c. Risk differences, Poor vs well-off childhood SES (and other exposures), by age group

Outcome	Exposure	age_group	contrast	RD pp (95% CI)
Cognitive impairment	Childhood SES	45-59	Poor - Pretty well-off	0.29 (-1.91 to 1.95)
Cognitive impairment	Childhood SES	60-74	Poor - Pretty well-off	-0.71 (-3.94 to 2.07)
Cognitive impairment	Childhood SES	75+	Poor - Pretty well-off	-8.83 (-16.21 to -1.38)
Cognitive impairment	Childhood health	45-59	Poor - Good	-0.72 (-2.39 to 1.02)
Cognitive impairment	Childhood health	60-74	Poor - Good	-0.12 (-2.54 to 2.64)
Cognitive impairment	Childhood health	75+	Poor - Good	-2.42 (-8.63 to 4.82)
Cognitive impairment	Childhood bedridden	45-59	Yes - No	0.56 (-0.9 to 2.26)
Cognitive impairment	Childhood bedridden	60-74	Yes - No	-0.62 (-3.65 to 2.95)
Cognitive impairment	Childhood bedridden	75+	Yes - No	-0.04 (-10.44 to 16.22)
Depressive symptoms	Childhood SES	45-59	Poor - Pretty well-off	4.32 (0.57 to 7.77)
Depressive symptoms	Childhood SES	60-74	Poor - Pretty well-off	6.2 (0.92 to 10.95)
Depressive symptoms	Childhood SES	75+	Poor - Pretty well-off	3.98 (-5.54 to 11.52)
Depressive symptoms	Childhood health	45-59	Poor - Good	5.0 (0.9 to 10.16)
Depressive symptoms	Childhood health	60-74	Poor - Good	6.82 (2.42 to 11.87)
Depressive symptoms	Childhood health	75+	Poor - Good	2.25 (-6.71 to 12.89)
Depressive symptoms	Childhood bedridden	45-59	Yes - No	3.91 (-1.01 to 8.94)
Depressive symptoms	Childhood bedridden	60-74	Yes - No	4.28 (0.19 to 8.1)
Depressive symptoms	Childhood bedridden	75+	Yes - No	6.73 (-6.68 to 27.94)

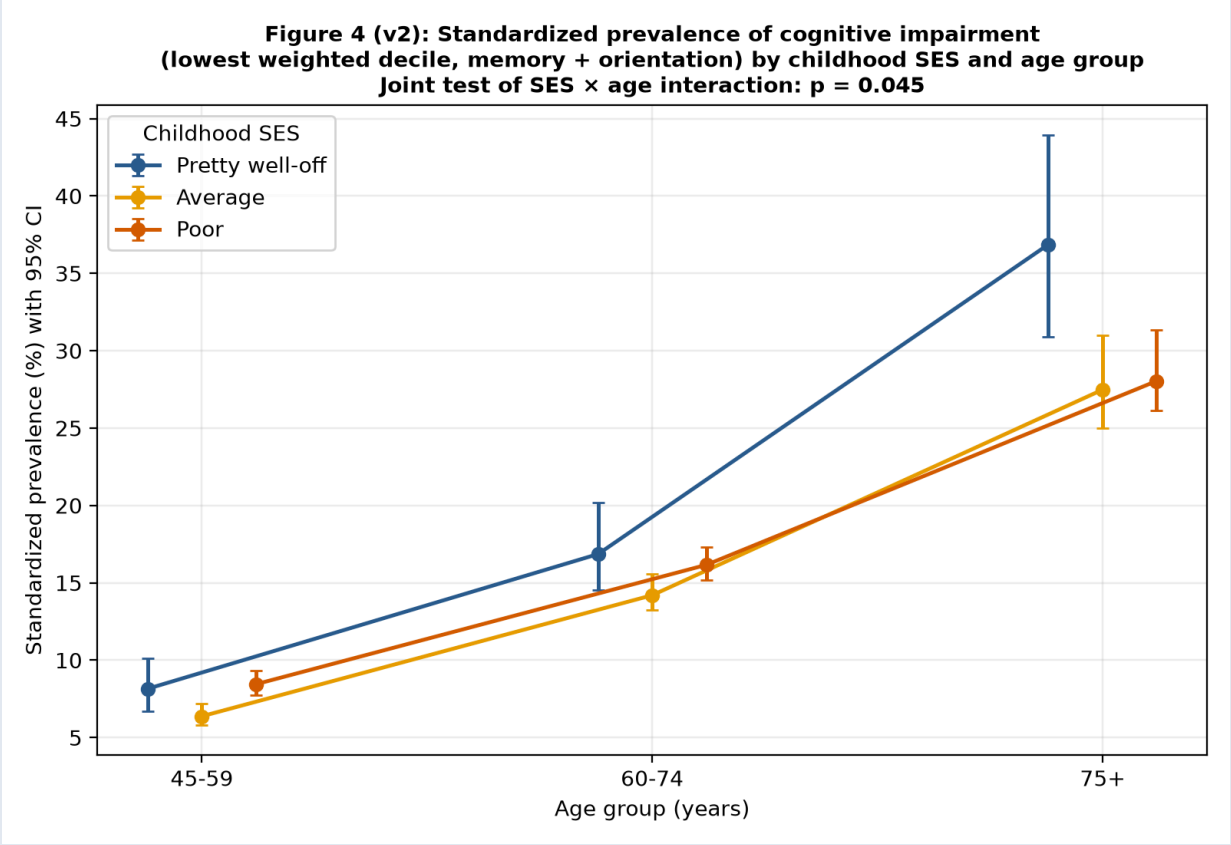


Figure 4 (v2): Standardized prevalence of cognitive impairment by childhood SES and age group.

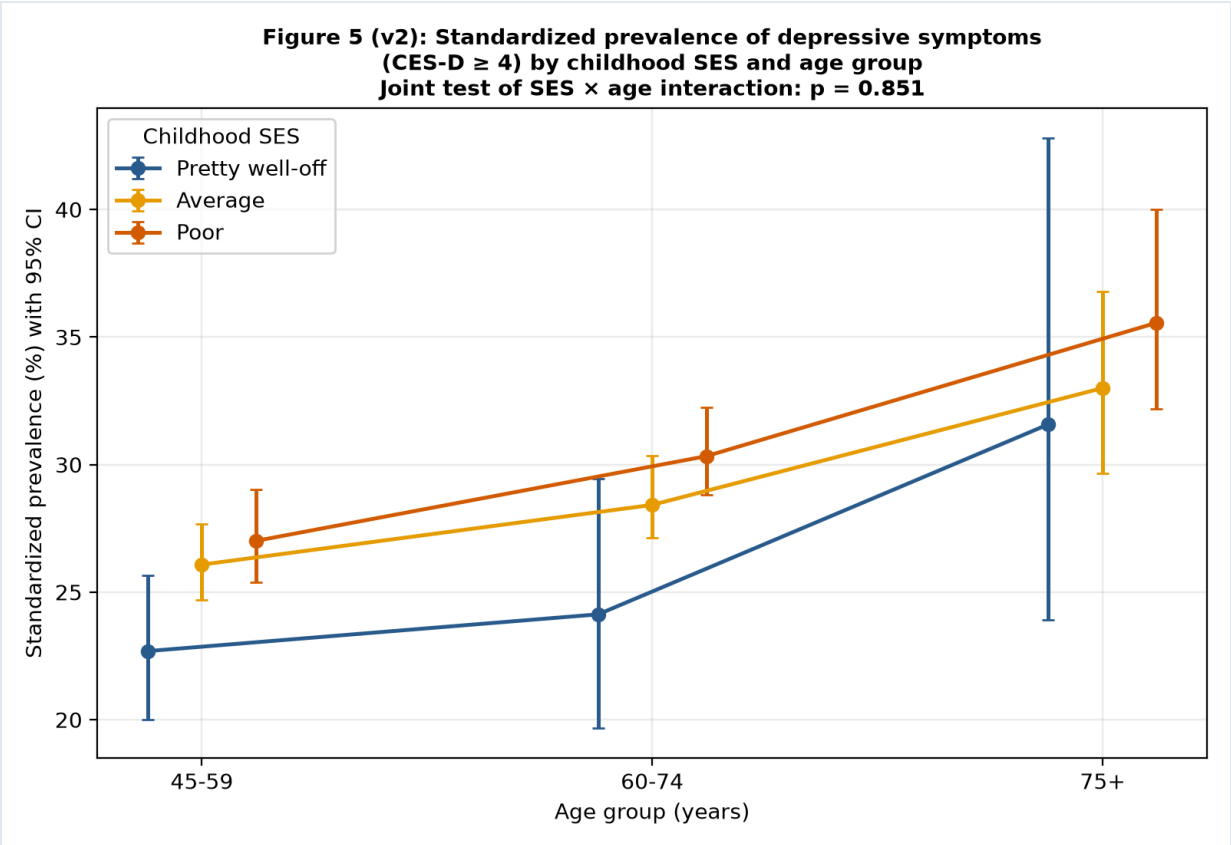


Figure 5 (v2): Standardized prevalence of depressive symptoms by childhood SES and age group.

- **Cognition:** the nominal interaction arises from the 75+ group, where the well-off group's standardized prevalence (36.9%, 30.9–44.0) is *higher* than the Poor group's (28.0%, 26.1–31.4). This is opposite to the hypothesized direction. It rests on 5,257 people, and 19.9% of the 75+ group lack a score. We treat it as exploratory.
- **Depression:** no interaction on the relative scale (p = 0.85). The absolute gap for poor SES is 4.3 pp (0.6 to 7.8) at 45–59 and 6.2 pp (0.9 to 11.0) at 60–74. At 75+ it is 4.0 pp with a CI spanning zero (−5.5 to 11.5), which is imprecise rather than evidence of convergence.
- **Hypothesis 4 is therefore not supported.** Explanations such as survivor selection are speculative. Age and birth cohort cannot be separated in a cross-section.

8. Sensitivity Analyses (Depression and Other Checks)

Analysis	Outcome	Term	Estimate (95% CI)	N
S4c SES incl. Varied category / M2	Cognitive impairment	v_poor	0.93 (0.82–1.06) p=0.279	57597
S4c SES incl. Varied category / M2	Cognitive impairment	v_var	1.39 (0.93–2.08) p=0.105	57597
S4c SES incl. Varied category / M2	Depressive symptoms	v_poor	1.20 (1.03–1.41) p=0.021	62429
S4c SES incl. Varied category / M2	Depressive symptoms	v_var	1.18 (0.89–1.57) p=0.253	62429
S4d Exclude age>=100 / M2	Cognitive impairment	ses_poor	0.93 (0.82–1.05) p=0.242	57226
S4d Exclude age>=100 / M2	Cognitive impairment	hlth_poor	0.95 (0.82–1.10) p=0.492	57226
S4d Exclude age>=100 / M2	Depressive symptoms	ses_poor	1.21 (1.03–1.41) p=0.019	62011
S4d Exclude age>=100 / M2	Depressive symptoms	hlth_poor	1.20 (1.07–1.34) p=0.002	62011
S4e Depression: modified Poisson PR / M2	CES-D>=4	hlth_poor	1.20 (1.07–1.34) p=0.002	62047
S4e Depression: logistic OR / M2	CES-D>=4	hlth_poor	1.33 (1.11–1.58) p=0.002	62047
S4e Depression: continuous CES-D / M2	CES-D score 0-10 (beta)	hlth_poor	0.36 (0.25–0.48) p=0.000	62047
S4e Depression: continuous CES-D / M2	CES-D score 0-10 (beta)	ses_poor	0.32 (0.15–0.50) p=0.000	62047
S4e Childhood health 5-level, per-level trend / M2	CES-D>=4	h5	1.12 (1.07–1.16) p=0.000	62047
S4e Childhood health 5-level, per-level trend / M2	Cognitive impairment	h5	1.03 (0.98–1.09) p=0.260	57245
S4f Sex-stratified (mutually adjusted, M2): Female	Cognitive impairment	ses_poor	0.97 (0.84–1.13) p=0.712	30115
S4f Sex-stratified (mutually adjusted, M2): Female	Cognitive impairment	hlth_poor	0.98 (0.84–1.15) p=0.842	30115
S4f Sex-stratified (mutually adjusted, M2): Female	Cognitive impairment	bed_yes	1.09 (0.87–1.38) p=0.443	30115
S4f Sex-stratified (mutually adjusted, M2): Female	Depressive symptoms	ses_poor	1.23 (1.01–1.49) p=0.037	33226
S4f Sex-stratified (mutually adjusted, M2): Female	Depressive symptoms	hlth_poor	1.21 (1.09–1.35) p=0.001	33226
S4f Sex-stratified (mutually adjusted, M2): Female	Depressive symptoms	bed_yes	1.10 (0.99–1.23) p=0.069	33226
S4f Sex-stratified (mutually adjusted, M2): Male	Cognitive impairment	ses_poor	0.83 (0.65–1.07) p=0.147	27130
S4f Sex-stratified (mutually adjusted, M2): Male	Cognitive impairment	hlth_poor	0.89 (0.72–1.09) p=0.248	27130
S4f Sex-stratified (mutually adjusted, M2): Male	Cognitive impairment	bed_yes	0.86 (0.67–1.10) p=0.237	27130
S4f Sex-stratified (mutually adjusted, M2): Male	Depressive symptoms	ses_poor	1.14 (0.96–1.35) p=0.146	28821
S4f Sex-stratified (mutually adjusted, M2): Male	Depressive symptoms	hlth_poor	1.15 (0.99–1.34) p=0.075	28821
S4f Sex-stratified (mutually adjusted, M2): Male	Depressive symptoms	bed_yes	1.13 (0.94–1.35) p=0.198	28821

- The logistic OR for childhood health and depression (1.33) exceeds the modified-Poisson PR (1.20). The outcome is common (28%), so the OR overstates the relative prevalence.
- Continuous CES-D: childhood health +0.37 points (0.25–0.48); poor SES +0.32 (0.15–0.50).
- Five-level childhood health: PR 1.12 per level (1.07–1.16) for depression; 1.03 (0.98–1.09) for cognition.
- Sex-stratified depression estimates are similar in direction (women: health PR 1.21, 1.09–1.35; men: 1.15, 0.99–1.34).
- Excluding ages ≥100, or including the "Varied" SES category, leaves the conclusions unchanged.

9. Hypotheses: Updated Verdicts

Hypothesis	Cognition	Depression
H1 Childhood SES	Associated with impairment before adult SES adjustment (PR 1.36, M1); no association after education/MPCE on the primary outcome. Persists on the broader battery (PR 1.29).	Supported (PR 1.21, 1.03–1.42, M2).
H2 Childhood health	Not supported on memory + orientation (PR 0.95). Supported on the broader battery (PR 1.22).	Supported (PR 1.20, 1.07–1.34, M2; persists in M3).
H3 Bedridden ≥1 month	Not supported (PR 1.00).	Modestly supported (PR 1.16, 1.05–1.28; lower CI near 1 under mutual adjustment).
H4 Age heterogeneity	Not supported robustly (one nominal p = 0.045, in the opposite direction at 75+).	Not supported (all p > 0.7).
H5 Attenuation after adult SES	Attenuation seen for SES, but the primary outcome is itself education-linked, so no pathway interpretation.	Partial attenuation for SES (~30% of the log-PR); little for childhood health.

10. Study Limitations

1. **Cross-sectional design.** No temporal ordering of outcome and measured adult covariates. Education and MPCE are adjusted for but cannot be treated as mediators.
2. **Cognitive measurement.** Every threshold-based measure is related to education (Figure A1), and conclusions differ by outcome definition (Section 4).
3. **Retrospective exposures.** Childhood conditions were reported at ages 45–116. Recall error and mood-congruent reporting are plausible, especially for depression, where both exposure and outcome are self-reported.
4. **Missing cognition scores** are selective by age, education, sex and childhood SES. IPW assumes missing-at-random.
5. **Survey design.** `ssuid` is the only clustering variable available.
6. **Survivor selection** in the oldest ages may mask or reverse gradients.
7. **Multiple testing.** About 40 exposure × model × outcome estimates are reported without formal correction.
8. **Residual confounding** by factors such as childhood nutrition, parental education and migration is possible.

11. Conclusions with Empirical Evidence

1. **Depressive symptoms are the more robust finding.** Poor childhood health (PR 1.20, 95% CI 1.07–1.34), poor childhood SES (1.21, 1.03–1.42) and bedridden illness (1.16, 1.05–1.28) are each associated with a higher prevalence of CES-D ≥4. These estimates persist after adult education and household expenditure. They also hold in mutually adjusted models (1.19, 1.18, 1.12), on the continuous CES-D score, in sex-stratified analyses, and per level of childhood health (1.12). The cumulative-disadvantage index gives PR 1.11 per adversity (1.07–1.16).
2. **The cognition result is not robust.** On the memory + orientation outcome, the association with poor childhood SES (PR 1.36) disappears after adult education and expenditure (PR 0.93, 0.82–1.05). Childhood health and bedridden history show no association (PRs ≈ 0.95–1.00). On the broader skip-aware battery, the associations remain (PR 1.29 and 1.22). The conclusion therefore depends on which cognitive tasks are used.
3. **The v1 claim that adult schooling "fully mediates" the SES–cognition link was not supported.** The outcome itself is education-linked, and mediation cannot be assessed here.
4. **There is no robust evidence of age heterogeneity.** Joint tests were null except one nominal result (p = 0.045, cognition) in the opposite direction at 75+.
5. **Informative missingness did not change the estimates** under IPW (Poor SES, M2: 0.928 both ways), within the missing-at-random assumption.
6. **Implication:** the evidence favours early-life health and socioeconomic disadvantage as correlates of depressive symptoms in older Indian adults. For cognition, an education-adjusted memory + orientation outcome shows no such pattern, and the broader battery does. We recommend longitudinal follow-up (LASI Wave 2 onwards) and education-fair cognitive measures before drawing policy conclusions about cognition.

Appendix: reproducibility

Scripts: `code/07_build_dataset_v2.py`, `code/survey_utils.py`, `code/08_models_v2.py`, `code/09_forest_plot_v2.py`, `code/10_generate_report_v2.py`. Tables: `outputs/tables_v2/`. Figures: `outputs/figures_v2/`.